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Letter to the Editor

Prevalence of positive HIV, HBV, HCV and treponemal tests in blood donors in a rural hospital in southern Ethiopia

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Blood transfusion is common practice in tropical countries to treat endemic diseases such as malaria or chronic anemia from multiple causes. Human immunodeficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV) and syphilis represent major public health problems throughout the world. Ethiopia is among the countries where HIV and HBV infections are particularly prevalent. Screening of these blood-borne infections is crucial to prevent their transmission, and to diagnose and treat infected individuals [1]. Furthermore, convenience samples of volunteer blood donors provide valuable information about population frequencies of blood-borne infections; on the other hand, they may be non-representative of the general population, as indicated by several cross-sectional reports from Ethiopia [2–5]. The aim of this study was to determine the prevalence of HIV, HBV, HCV and syphilis among blood donors attended in a rural hospital in Southern Ethiopia.

A retrospective cross-sectional study was conducted from September 2007 to September 2012 in Gambo Rural Hospital (GRH), a 135-bed center located in the West-Arsi zone, 250 km south of Addis Ababa, Ethiopia. The hospital does not have a Blood Bank Unit, so blood donations to demand are common practice. All blood donors need to be older than 16 years of age and are usually relatives of patients with severe anemia, not volunteers. A daily registration system is performed in the hospital for all blood donations and entered in a specific logbook. During the course of our study period, there was no change in donor characteristics. In most cases, each donor made only one donation. Rapid immunochromatographic tests in capillary blood are used for detection of HIV-1 and HIV-2 (Colloidal Gold; KHB Shanghai, Kehua Bio-engineering Co., Ltd., China), HCV antibodies (HCV Rapid Hepatitis C Virus Test (Strip), Zhejiang Orient Gene Biotech Co., Ltd., Zhejiang, China) and hepatitis B surface antigen (HBsAg) (SD BIOLINE HBsAg Fast test; Gyeonggi-do, Korea), while the rapid plasma reagin (RPR) test (Wondfo One Step; Guangzhou Wondfo Biotech Co., Ltd., Guangzhou, China) is used to identify treponemal infection. Patient records reviewed for the study were all anonymous, and Ethics Committee approval was obtained from the local Research and Publication Committee of the GRH. The patients with HIV infection were sent to antiretroviral clinic and were attended according National Protocols of antiretroviral treatment. The patients testing positive for HCV received counselling about the infection, but the treatment and HCV viral load were not available.

Table 1
HIV and HCV serology, HBV antigen and RPR results in capillary blood from Ethiopian blood donors.

	N (%)
Anti HIV antibody	2605
Positive	19 (0.7)
Negative	2586 (99.3)
HBsAg	2128
Positive	129 (6.1)
Negative	1999 (93.9)
Anti HCV antibody	2223
Positive	44 (2.0)
Negative	2179 (98.0)
RPR	2537
Positive	22 (0.9)
Negative	2515 (99.1)

Of the 2606 individuals screened, a total of 19, 129, 44 and 22 patients had positive HIV, HBV, HCV or RPR results, respectively. Overall prevalences were 0.7% (95% CI 0.5–1.1) for HIV antibodies, 6.1% (95% CI 5.1–7.1) for HBsAg, 2.0% (95% CI 1.5–2.6) for HCV antibodies and 0.9% (95% CI 0.7–1.3) for positive RPR (Table 1).

When prevalence of infections was analyzed by sex, only HBV infection was more common in men than in women (6.6% vs. 4.1%, $p = 0.05$). With respect to age, only HIV infection was more common among those aged 25–34 years, as compared to younger subjects (1.1% vs. 0.3%, $p = 0.03$) (Fig. 1).

Among HIV-infected patients, prevalence of positive HBsAg and treponemal infection was significantly greater than in HIV-negative patients (36.4% vs. 5.9%, ($p < 0.001$) and 8.3% vs. 0.4% ($p = 0.005$), respectively). In HBsAg-positive patients, prevalence of HIV and HCV antibodies was significantly higher than in HBsAg-negative patients (3.1% vs. 0.4% ($p < 0.001$) and 12.3% vs. 1.4% ($p < 0.001$), respectively). Prevalence of HBsAg was significantly higher in HCV-positive than in HCV-negative individuals (33.3% vs. 4.9% $p < 0.001$). Finally, HIV infection was more common among patients with treponemal infection as compared with RPR negative (4.5% vs. 0.4% $p = 0.005$) (Fig. 1).

The frequency of co-infections found in our series is in line with results from other studies [4,5]. Association between blood-borne (HIV, HBV and HCV) and sexually transmitted (HIV, HBV and syphilis) pathologies indicate that both routes are common means of infection in Ethiopia. The association between HBV, HCV and syphilis with HIV infection is noteworthy, given that immune depression complicates the outcome of those infections. As such, with HIV co-infection there is a greater risk for HBV or HCV advanced liver diseases, or for central nervous system treponemal disease. The same is true for the association between HBV and HCV

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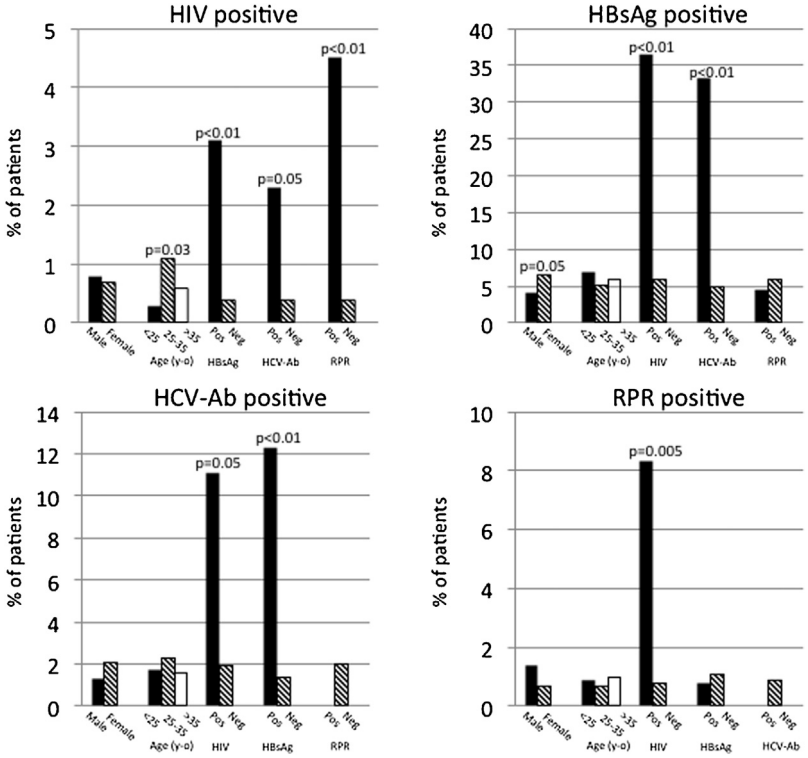


Fig. 1. HIV and HCV serology, HBV antigen and RPR results in capillary blood from Ethiopian blood donors analized by age, gender and coinfections.

infections, a recognized risk factor for faster progression of liver fibrosis.

Recently, the status of transfusion service has changed at GRH. The National Blood Transfusion Services (NBTS) was first established in 1969 by the Ethiopian Red Cross society; transferred in 2004 to the Federal Ministry of Health Ethiopia, the NBTS is responsible for screening the collected blood for transfusion–transmissible infections (HIV, HBV, HCV and syphilis) according to quality standards. However, only in June 2015 was GRH included in the program, after making several improvements.

The results of this study reinforce the need for mandatory screening for transfusion-transmissible infections (HIV, HBV, HCV and syphilis) in blood donations in order to prevent transmission of these blood-borne conditions in rural area were a NBTS is not improved. When the NBTS can ensure the correct screening of the collected blood for transfusion in all private and public centers, the transmission can be stopped.

Conflict of interest

There are no conflicts of interest, but an absence of any relationship or any degree of conflicting or dual interest, financial or of any other nature that may affect professional judgment in relation to this article.

Ethical approval

Ethics Committee approval was obtained from the local Research and Publication Committee of the GRH, and Health Unit and Ethical Review Committee of the Ethiopian Catholic Secretary.

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